Grazioso Salvare README

About the Project

This project is designed to filter breeds of dogs from a database of animals from the Austin Animal Shelter and display graphs to help the user interpret the filtered information.

Motivation

This project exists so that the rescue-animal training company Gravioso Salvare can identify and categorize dogs that are appropriate for different types of training. Specifically, Gravioso Salvare wants to find dogs suited for Water Rescue, Mountain or Wilderness Rescue, and Disaster Rescue or Individual Tracking.

Getting Started

To get a local copy up and running, follow these steps:

1. Import the Austin Animal Shelter data set in MongoDB by running
   ```
   cd ./datasets
   mongoimport --type=csv --headerline --db aac --collection animals --drop ./aac_shelter_outcomes.csv
   ```
2. Create a new user account for the database by running
   ```
   mongosh
   use admin
   db.createUser( { user: "aacuser", pwd:  passwordPrompt(), roles: [ { role: "readWrite", db: "aac" } ] } )
   ```
3. Run the ProjectTwoDashboard.ipynb file in Jupyter Lab
4. Connect to the Dash dashboard using the link output by Jupyter Lab

Installation

You will need the Austin Animal Shelter Outcomes dataset. Note, a working version of this dataset is provided. If you want to use a newer version, ensure the expected fields are present:
https://data.austintexas.gov/Health-and-Community-Services/Austin-Animal-Center-Outcomes/gsvs-ypi7/about_data
You need the Python interpreter to execute the Python code:
https://www.python.org/
You also need an installation of Jupyter Lab to run the Python code:
https://jupyter.org/install
You need MongoDB Community Server to access the animal shelter database. The associated service may need to be started manually:
https://www.mongodb.com/try/download/community
You need the MongoDB Command Line Database Tools to use mongoimport:
https://www.mongodb.com/try/download/database-tools
You need the Mongo Shell to use the mongosh command:
https://www.mongodb.com/try/download/shell
PyMongo is used to work with MongoDB in Python:
https://pypi.org/project/pymongo/
Dash is used to build the web framework for the Python code:
https://dash.plotly.com/

Why MongoDB

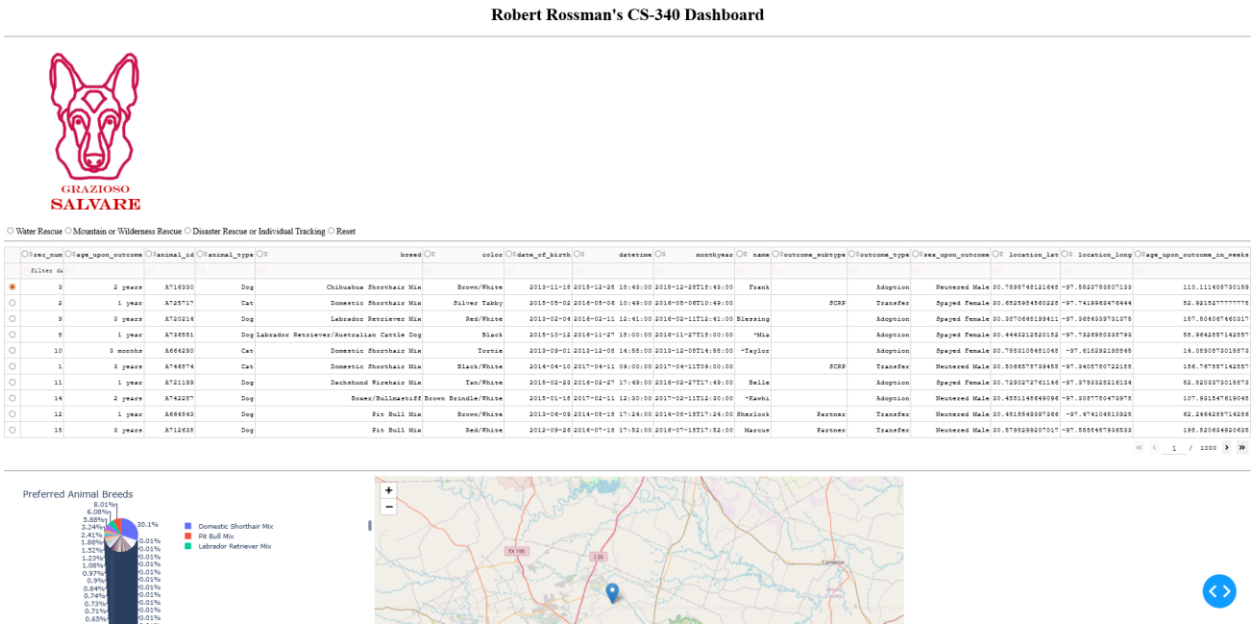
MongoDB was used as the model component of the development because it is well-suited for working with and manipulating large databases. Paired with PyMongo, MongoDB is also able to be used with Python code that runs functions on the animal shelter database.

Why the Dash Framework

The Dash framework works with Python and MongoDB to build a web framework. Dash provides the view structure for the web application as it displays HTML components that show the entries in the database and graphs of the data. Along with this, Dash provides the controller structure for the web application as it allows the user to interact with it on the client level using things like radio buttons to manipulate the information that is displayed.

Functionality Screenshots

Starting state of the dashboard:



"Water Rescue" data filter applied:

Robert Rossman's CS-340 Dashboard

GRAZIOSO SALVARE

◉ Water Rescue ○ Mountain or Wilderness Rescue ○ Disaster Rescue or Individual Tracking ○ Reset

	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome
	filter data..												
◉	549	6 years	A749092	Dog	Labrador Retriever	Brown	2011-01-12	2017-05-13 17:22:00	2017-05-13T17:22:00	Lucas		Return to Owner	Neutered Male
○	554	4 years	A687175	Dog	Labrador Retriever	Black/White	2010-09-02	2014-11-11 13:16:00	2014-11-11T13:16:00	*Carla		Adoption	Spayed Female
○	1128	10 months	A731765	Dog	Labrador Retriever	Black/White	2015-09-27	2016-08-01 19:14:00	2016-08-01T19:14:00			Adoption	Neutered Male
○	1190	1 month	A700479	Dog	Labrador Retriever	Brown/White	2015-03-03	2015-05-03 19:41:00	2015-05-03T19:41:00	Quinn	Offsite	Adoption	Spayed Female
○	1746	9 years	A726075	Dog	Labrador Retriever	Chocolate	2008-05-06	2017-05-04 09:53:00	2017-05-04T09:53:00	Henry		Return to Owner	Neutered Male
○	2028	2 months	A741619	Dog	Labrador Retriever	White/Black	2016-10-21	2017-01-09 14:53:00	2017-01-09T14:53:00	Maximus		Adoption	Intact Male
○	3247	5 years	A716279	Dog	Labrador Retriever	Chocolate	2010-11-17	2015-12-24 13:19:00	2015-12-24T13:19:00	*Sampson		Adoption	Neutered Male
○	3363	7 years	A577561	Dog	Labrador Retriever	Yellow	2008-06-26	2016-01-13 19:11:00	2016-01-13T19:11:00	Goyo		Return to Owner	Intact Male
○	3574	10 years	A698205	Dog	Labrador Retriever	Yellow	2005-03-07	2015-03-07 14:22:00	2015-03-07T14:22:00	Zeke		Return to Owner	Intact Male
○	4276	4 years	A683067	Dog	Labrador Retriever	Cream	2010-07-06	2014-07-12 18:40:00	2014-07-12T18:40:00	Gusto		Adoption	Intact Male

Preferred Animal Breeds





“Mountain or Wilderness Rescue” data filter applied:

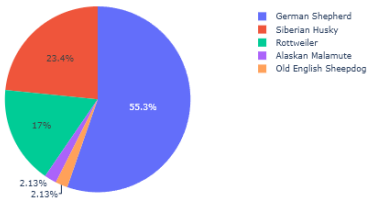
Robert Rossman's CS-340 Dashboard

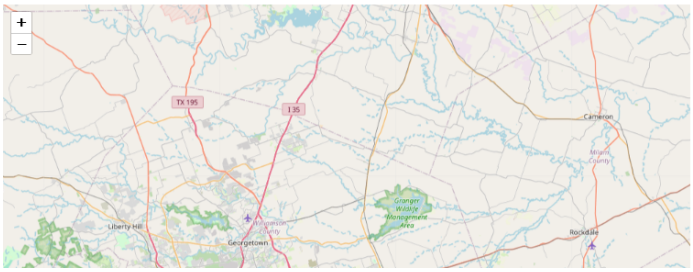
GRAZIOSO SALVARE

○ Water Rescue ◉ Mountain or Wilderness Rescue ○ Disaster Rescue or Individual Tracking ○ Reset

	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type
	filter data...											
◉	304	13 years	A358879	Dog	German Shepherd	Brown/Black	2002-08-13	2015-09-15 16:22:00	2015-09-15T16:22:00	Simba	Suffering	Euthanasia
○	400	11 months	A681819	Dog	German Shepherd	Black	2013-07-15	2014-07-02 16:40:00	2014-07-02T16:40:00	Ashes		Return to Owner
○	434	2 years	A726876	Dog	Rottweiler	Black/Brown	2014-05-15	2016-09-22 19:08:00	2016-09-22T19:08:00	*Reuben		Adoption
○	551	4 years	A662100	Dog	German Shepherd	Brown/Black	2010-03-01	2015-02-19 12:59:00	2015-02-19T12:59:00	Beaux	Partner	Transfer
○	1103	7 years	A738749	Dog	German Shepherd	Brown/Black	2009-11-20	2016-11-25 14:09:00	2016-11-25T14:09:00	Camelia		Adoption
○	1114	5 months	A736023	Dog	Siberian Husky	Red/White	2016-05-06	2016-10-11 13:04:00	2016-10-11T13:04:00	Benson		Return to Owner
○	1172	11 months	A762987	Dog	Siberian Husky	White/Brown	2017-01-19	2018-01-04 18:55:00	2018-01-04T18:55:00	Jack		Return to Owner
○	1364	5 months	A760122	Dog	German Shepherd	Black/Brown	2017-03-13	2017-09-04 12:11:00	2017-09-04T12:11:00	Zeus		Return to Owner
○	1488	2 months	A692264	Dog	German Shepherd	Black/Tan	2014-03-16	2014-05-31 16:31:00	2014-05-31T16:31:00		Partner	Transfer
○	1541	2 years	A743624	Dog	German Shepherd	Red/White	2015-02-13	2017-02-14 00:00:00	2017-02-14T00:00:00	Gavin	Partner	Transfer

Preferred Animal Breeds





“Disaster Rescue or Individual Tracking” data filter applied:

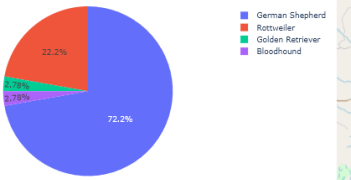
Robert Rossman's CS-340 Dashboard

GRAZIOSO SALVARE

○ Water Rescue ○ Mountain or Wilderness Rescue ◉ Disaster Rescue or Individual Tracking ○ Reset

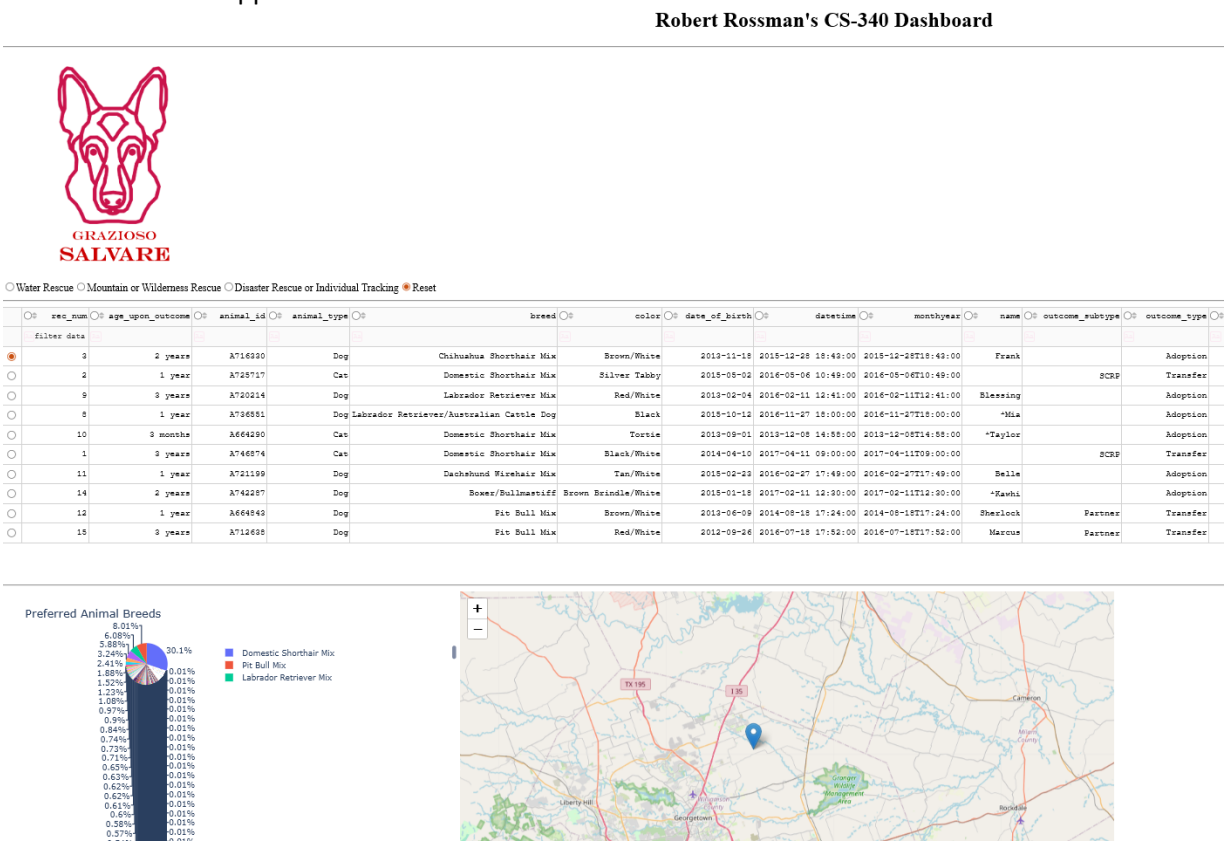
	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome
	filter data...												
◉	234	2 years	A734683	Dog	Golden Retriever	Red	2014-09-11	2016-09-15 11:24:00	2016-09-15T11:24:00	Sapphire	Partner	Transfer	Spayed Female
○	304	12 years	A358879	Dog	German Shepherd	Brown/Black	2002-08-13	2015-09-15 16:22:00	2015-09-15T16:22:00	Simba	Suffering	Euthanasia	Neutered Male
○	400	11 months	A681819	Dog	German Shepherd	Black	2013-07-15	2014-07-02 16:40:00	2014-07-02T16:40:00	Ashes		Return to Owner	Spayed Female
○	434	2 years	A726576	Dog	Rottweiler	Black/Brown	2014-05-15	2016-09-22 19:05:00	2016-09-22T19:05:00	*Reuben		Adoption	Neutered Male
○	551	4 years	A662100	Dog	German Shepherd	Brown/Black	2010-02-01	2015-02-19 12:59:00	2015-02-19T12:59:00	Beaux	Partner	Transfer	Neutered Male
○	1103	7 years	A738749	Dog	German Shepherd	Brown/Black	2009-11-20	2016-11-25 14:09:00	2016-11-25T14:09:00	Camelia		Adoption	Spayed Female
○	1364	5 months	A760132	Dog	German Shepherd	Black/Brown	2017-03-13	2017-09-04 12:11:00	2017-09-04T12:11:00	Zeus		Return to Owner	Neutered Male
○	1488	2 months	A680264	Dog	German Shepherd	Black/Tan	2014-02-16	2014-05-21 16:21:00	2014-05-21T16:21:00		Partner	Transfer	Intact Female
○	1541	2 years	A743624	Dog	German Shepherd	Red/White	2015-02-13	2017-02-14 00:00:00	2017-02-14T00:00:00	Gavin	Partner	Transfer	Neutered Male
○	1544	2 years	A746092	Dog	Rottweiler	Black/Tan	2015-03-28	2017-04-01 13:45:00	2017-04-01T13:45:00	Nibbles	Partner	Transfer	Neutered Male

Preferred Animal Breeds





“Reset” data filter applied:



	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type
	filter data											
◉	3	2 years	A716320	Dog	Chihuahua Shorthair Mix	Brown/White	2013-11-15	2015-12-28 18:43:00	2015-12-28T18:43:00	Frank		Adoption
○	2	1 year	A725717	Cat	Domestic Shorthair Mix	Silver Tabby	2015-05-02	2016-05-06 10:49:00	2016-05-06T10:49:00		SCRP	Transfer
○	9	3 years	A720214	Dog	Labrador Retriever Mix	Red/White	2013-02-04	2016-02-11 12:41:00	2016-02-11T12:41:00	Blessing		Adoption
○	5	1 year	A736551	Dog	Labrador Retriever/Australian Cattle Dog	Black	2015-10-12	2016-11-27 18:00:00	2016-11-27T18:00:00	*Mia		Adoption
○	10	3 months	A664290	Cat	Domestic Shorthair Mix	Tortie	2013-09-01	2013-12-05 14:58:00	2013-12-05T14:58:00	*Taylor		Adoption
○	1	3 years	A746574	Cat	Domestic Shorthair Mix	Black/White	2014-04-10	2017-04-11 09:00:00	2017-04-11T09:00:00		SCRP	Transfer
○	11	1 year	A721199	Dog	Dachshund Wirehair Mix	Tan/White	2015-02-23	2016-02-27 17:49:00	2016-02-27T17:49:00	Belle		Adoption
○	14	2 years	A742257	Dog	Boxer/Bullmastiff	Brown Brindle/White	2015-01-18	2017-02-11 12:30:00	2017-02-11T12:30:00	*Kawhi		Adoption
○	12	1 year	A664843	Dog	Pit Bull Mix	Brown/White	2013-06-09	2014-08-18 17:24:00	2014-08-18T17:24:00	Sherlock	Partner	Transfer
○	15	3 years	A712638	Dog	Pit Bull Mix	Red/White	2012-09-26	2016-07-18 17:52:00	2016-07-18T17:52:00	Marcus	Partner	Transfer

Steps Taken

To complete this project, I first imported a previously written CRUD python module that allowed me to connect to the animal shelter database and read and filter its entries. From there, I built the layout of the app, including a unique identifier, the Grazioso Salvare logo with a URL anchor tag, and a DataTable with selectable columns and rows.

Importantly, I also added radio buttons corresponding to “Water Rescue,” “Mountain or Wilderness Rescue,” “Disaster Rescue or Individual Tracking,” and “Reset” data filters. I made it so that selecting these radio buttons uses a callback to change the filter currently being used which then updates the dashboard accordingly. A pie chart is also displayed which uses callbacks to update the breeds of the animals it shows according to the current filter. Lastly, a map is displayed which uses callbacks to update which animal it displays the location of based on which animal is currently selected in the DataTable.

Challenges

Most of my challenges in this project came from trying to get the `update_dashboard` callback to work. Aside from initially having some syntax errors, my major problem came from me neglecting to include the function `df.drop(columns=['_id'], inplace=True)` to clean up the Mongo `_id` field.

I also had some difficulty with displaying the Grazioso Salvare logo but found that I simply needed to add the code to display the image into the `app.layout`. After discovering this, it was trivial to add the URL anchor using `href` and even to scale down the image using `width` and `height`.

Contact
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